

Lecture 1: Difference Equation

1-st order difference: $\Delta y_t = y_t - y_{t-1}$

2-nd order difference: $\Delta^2 y_t = \Delta(\Delta y_t) = \Delta(y_t - y_{t-1}) = \Delta y_t - \Delta y_{t-1} = y_t - y_{t-1} - (y_{t-1} - y_{t-2})$
 $= y_t - 2y_{t-1} + y_{t-2}$

Period: an interval of time $\Rightarrow y_t$ is the y for t^{th} interval of time, not for a point of time.
 * t 是一个时间段概念, 不是时间点概念

$\Delta y_t = y_t - y_{t-1}$ is the change from $(t-1)^{\text{th}}$ time interval to t^{th} interval, not a derivative at a point of time t .

* Δy_t 是变化量, 并不是在 t 点的微分(变化率)

$$\text{Derivative} = \dot{y}_t = \frac{\Delta y_t}{\Delta t} = \frac{y_t - y_{t-1}}{1} = \frac{y_t}{y_{t-1}} - 1 \approx \log\left(\frac{y_t}{y_{t-1}}\right)$$

When $\frac{y_t}{y_{t-1}}$ is small, $\log\left(1 + \frac{y_t}{y_{t-1}} - 1\right)$

§ Solve a first-order difference equation $\Leftrightarrow$ Rewrite y_t as a function of t . $\approx \frac{y_t}{y_{t-1}} - 1 = \frac{y_t - y_{t-1}}{y_{t-1}}$

eg: $y_t + a y_{t-1} = C$

① Brutal force: iterative method, $y_t = -a y_{t-1} + C$
 $y_{t-1} = -a y_{t-2} + C \quad \left| \Rightarrow y_t = a^2 y_{t-2} - aC + C, \right.$
 $y_{t-2} = -a y_{t-3} + C \quad \left| y_{t-2} = -a y_{t-3} + C \right.$

$\Rightarrow y_t = -a^3 y_{t-3} + a^2 C - aC + C \dots \Rightarrow y_t = (-a)^t y_0 + C \sum_{j=0}^{t-1} (-a)^j$
 $\sum_{j=0}^{t-1} (-a)^j = \frac{1 - (-a)^t}{1 + a}$ if $a \neq -1$

$\Rightarrow y_t = (-a)^t \left(y_0 - \frac{C}{1+a}\right) + \frac{C}{1+a}$, if $a \neq -1$

$\Rightarrow y_t = A(-a)^t + \frac{C}{1+a}$, if $a \neq -1$

* If $a = -1$, $\sum_{j=0}^{t-1} (-a)^j = \sum_{j=0}^{t-1} 1 = t \Rightarrow y_t = y_0 + Ct$

$y_t = \begin{cases} A(-a)^t + \frac{C}{1+a}, & \text{if } a \neq -1 \\ Ct + y_0, & \text{if } a = -1 \end{cases}$, where $A = y_0 - \frac{C}{1+a}$

② General method

Since the system is linear, we can separate: ① the part driven by constant C

② the part comes from the dynamics

* why $y_t = y_0^p + y_t^c$ is valid? Define linear operator $L(\cdot) = (\cdot)_t + a(\cdot)_{t-1}$

Then $y_t + a y_{t-1} = C \Rightarrow L(y) = C$ (2)

If y^p is any particular solution that $L(y^p) = C$

If y^c is any solution of the homogeneous equation $L(y^c) = 0$ (complementary solution)

Then by linearity: $L(y^p + y^c) = L(y^p) + L(y^c) = C + 0 = C$

$\Rightarrow y^p + y^c$ solves the equation $y_t + a y_{t-1} = C$

* All solutions = one particular solution + all homogeneous solution.

Now we solve the difference equation with general method.

① Since exponential sequences are eigenfunctions of the lag operator, try $y_t = A b^t$

$\Rightarrow A b^t + a A b^{t-1} = 0 \Rightarrow b + a = 0 \Rightarrow b = -a \Rightarrow y_t = A(-a)^t$

② Since the forcing term is a constant C , we guess $y_t^p = k$

$\Rightarrow k + a k = C \Rightarrow k = \frac{C}{1+a}, a \neq -1$

If $a = -1$, $y_t - y_{t-1} = C \Rightarrow y_t = C t$

$\therefore y_t = y_c + y_p = \begin{cases} A(-a)^t + \frac{C}{1+a}, & a \neq -1 \\ A + C t, & a = -1 \end{cases}$

$y_0 = A(-a)^0 + \frac{C}{1+a} = A + \frac{C}{1+a} \Rightarrow A = y_0 - \frac{C}{1+a}, \text{ when } a \neq -1$

$y_0 = A \Rightarrow A = y_0, \text{ when } a = -1$

§ Dynamic stability of equilibrium:

$y_t + a y_{t-1} = C$ then steady state is $\bar{y} + a \bar{y} = C \Rightarrow \bar{y} = \frac{C}{1+a} (a \neq -1)$
($y_t = y_{t-1} = \bar{y}$)

Subtract steady state from the original equation $\Rightarrow y_t - \bar{y} + a(y_{t-1} - \bar{y}) = C - \bar{y} - a\bar{y}$

Define deviation from steady state as $\tilde{y}_t = y_t - \bar{y} \Rightarrow \tilde{y}_t + a \tilde{y}_{t-1} = C - \frac{C}{1+a} - \frac{Ca}{1+a}$

$\Rightarrow \tilde{y}_t = (-a) \tilde{y}_{t-1}$ * $b = -a$ controls convergence 收敛性

$\Rightarrow \tilde{y}_t = b^t \tilde{y}_0$ $\begin{cases} \text{If } |b| < 1, \text{ then } b^t \rightarrow 0, \tilde{y}_t \rightarrow 0, y_t \rightarrow \bar{y}, \text{ convergence} \\ \text{If } |b| > 1, \text{ then } b^t \rightarrow \infty, \tilde{y}_t \rightarrow \infty, \text{ divergence} \end{cases}$

Sign of b also controls oscillation 交替振荡性 / monotonicity 单调性 (3)

If $b > 0$, b^t keeps the same sign for $\forall t \Rightarrow \hat{y}_t$ keeps the same sign for $\forall t$, path is non-oscillatory (monotone towards / away from $\bar{y}$)

If $b < 0$, b^t alternates sign $\Rightarrow \hat{y}_t$ flips sign each period $\Rightarrow$ oscillate around $\bar{y}$.

Further:

If $b > 1$, y_t monotonically diverge from $\bar{y}$.

If $b = 1$, y_t keeps the same distance away from $\bar{y}$.

If $0 < b < 1$, y_t monotonically converge to $\bar{y}$.

If $b = 0$, y_t stays at steady state $\bar{y}$ always

If $-1 < b < 0$, y_t oscillate around and converge to $\bar{y}$.

If $b = -1$, y_t oscillate around $\bar{y}$ and keeps same absolute distance

If $b < -1$, y_t oscillate around and diverge from $\bar{y}$.

The role of A :

Since the general solution can be written as: $y_t = \bar{y} + Ab^t \Rightarrow A = y_0 - \bar{y}$ initial deviation
Scale effect: $|A| \uparrow \Rightarrow$ start farther from equilibrium $\Rightarrow$ bigger movements along path.

Mirror effect: $A > 0$: start above $\bar{y}$; $A < 0$: start below $\bar{y}$.

Example: Inventory model.

$$Q_{d,t} = \alpha - \beta P_t \dots \text{demand}$$

$$Q_{s,t} = -\gamma + \delta P_t \dots \text{supply}$$

$$P_{t+1} = P_t - \bar{\sigma}(Q_{s,t} - Q_{d,t})$$

$$\alpha, \beta, \gamma, \delta, \bar{\sigma} > 0$$

Solve for P_t 's path

$$Q_{s,t} - Q_{d,t} = -(\alpha + \gamma) + (\beta + \delta)P_t \quad \text{Plug in} \Rightarrow P_{t+1} = [1 - \bar{\sigma}(\beta + \delta)]P_t + \bar{\sigma}(\alpha + \gamma)$$

$$\text{At steady state } P_{t+1} = P_t = \bar{P} \Rightarrow \bar{P} = \frac{\alpha + \gamma}{\beta + \delta} \quad \text{Thus, the deviation dynamics is.}$$

$$P_t - \bar{P} = [1 - \bar{\sigma}(\beta + \delta)](P_{t-1} - \bar{P})$$

Stability condition tells us: $|1 - \bar{\sigma}(\beta + \delta)| < 1 \Rightarrow 0 < \bar{\sigma}(\beta + \delta) < 2$ system converges.

Oscillation condition tells us: $1 - \bar{\sigma}(\beta + \delta) < 0 \Rightarrow \bar{\sigma}(\beta + \delta) > 1$ creates oscillation

★ Economic intuition: if price adjustment is too aggressive $\bar{\sigma} \uparrow \Rightarrow$ creates oscillation and divergence.

If supply or demand elasticity in terms of price too high $\beta\pi/\delta\pi$

(4)

$\Rightarrow$ creates oscillation or even divergence.

§ Higher order difference equations.

2nd order linear difference equation with constant coefficients & constant term.

$$y_{t+2} + a_1 y_{t+1} + a_2 y_t = C$$

1° Particular solution y^p : the intertemporal equilibrium level.

Try a constant: $y^p = k \Rightarrow k(1 + a_1 + a_2) = C \Rightarrow k = \frac{C}{1 + a_1 + a_2}, 1 + a_1 + a_2 \neq 0$

Try a first-degree polynomial: $y^p = kt \Rightarrow k(t+2) + a_1 k(t+1) + a_2 kt = C$
 $\Rightarrow k[(1 + a_1 + a_2)t + (2 + a_1)] = C$ impose $1 + a_1 + a_2 = 0$
 $\Rightarrow k = \frac{C}{2 + a_1}, 2 + a_1 \neq 0$

Try a second-degree polynomial: $y^p = kt^2 \Rightarrow k(t+2)^2 + a_1 k(t+1)^2 + a_2 kt^2 = C$
 $\left. \begin{aligned} 1 + a_1 + a_2 &= 0 \\ 2 + a_1 &= 0 \end{aligned} \right\} \Rightarrow \left. \begin{aligned} a_1 &= -2 \\ a_2 &= 1 \end{aligned} \right\} \Rightarrow$

$$k(t^2 + 4t + 4) - 2k(t^2 + 2t + 1) + kt^2 = C \Rightarrow k(t^2 + 4t + 4 - 2t^2 - 4t - 2 + t^2) = C$$

$$\Rightarrow 2k = C \Rightarrow k = \frac{C}{2}$$

$$\therefore y^p = \begin{cases} \frac{C}{1 + a_1 + a_2}, & \text{if } 1 + a_1 + a_2 \neq 0 \\ \frac{C}{1 + a_1} t, & \text{if } 1 + a_1 + a_2 = 0, a_1 \neq -2 \\ \frac{C}{2} t^2, & \text{if } a_1 = -2, a_2 = 1 \end{cases}$$

2° Complementary solution y^c : deviation from the equilibrium in each period.

Look at the homogeneous equation: $y_{t+2} + a_1 y_{t+1} + a_2 y_t = 0$ try $y_t = Ab^t$

$$\Rightarrow b^2 + a_1 b + a_2 = 0$$

Case 1: Two distinct real root: $a_1^2 - 4a_2 > 0 \Rightarrow y^c = A_1 b_1^t + A_2 b_2^t$

$$b_1 = \frac{-a_1 + \sqrt{a_1^2 - 4a_2}}{2}, b_2 = \frac{-a_1 - \sqrt{a_1^2 - 4a_2}}{2}$$

Case 2: Repeated real root: $a_1^2 - 4a_2 = 0 \Rightarrow y^c = A_3 b^t + A_4 t b^t$

$$b = b_1 = b_2 = -\frac{a_1}{2}$$

Case 3: Complex root $a_1^2 - 4a_2 < 0 \Rightarrow b_{1,2} = h \pm vi$, where $h = -\frac{a_1}{2}, v = \frac{\sqrt{4a_2 - a_1^2}}{2}$

$$y^c = A_5 (h + vi)^t + A_6 (h - vi)^t$$

Using De Moivre's Theorem: $(h+vi)^t = R^t (\cos \theta t + i \sin \theta t)$

where $R = \sqrt{h^2 + v^2} = \sqrt{a_2}$, $\cos \theta = \frac{h}{R} = -\frac{a_1}{2\sqrt{a_2}}$, $\sin \theta = \frac{v}{R} = \sqrt{1 - \frac{a_1^2}{4a_2}}$

$$\Rightarrow y^c = R^t (A_5 \cos \theta t + A_6 \sin \theta t)$$

where $R = \sqrt{a_2}$, θ is defined as $\begin{cases} \cos \theta = -\frac{a_1}{2\sqrt{a_2}} \\ \sin \theta = \sqrt{1 - \frac{a_1^2}{4a_2}} \end{cases}$

* Convergence under higher-order difference equation

1° Distinct root: b_{dom} is the root with highest absolute value.

convergence requires $|b_{dom}| < 1$

2° Repeated root: $|b| < 1 \Rightarrow$ converges

3° Complex root: $\begin{cases} R > 1: \text{explosive oscillations} \\ R < 1: \text{damped oscillations} \end{cases}$

Simultaneous difference equation (state-space form)

$$\text{let } x_t = y_{t+1} \Rightarrow \begin{cases} x_{t+1} + a_1 x_t + a_2 y_t = c \\ x_t = y_{t+1} \end{cases}$$

$$\Rightarrow \begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix} = \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_t \\ y_t \end{pmatrix} + \begin{pmatrix} c \\ 0 \end{pmatrix}$$

Generalize: $X_{t+1} = AX_t + B$ where $X_t = (x_{1t}, x_{2t}, \dots, x_{nt})$ is state space

A is a constant matrix with a_{ij} as elements, B is an array with $B_i, i=1, \dots, n$ as elements

$$\text{In this example: } X_{t+1} = \begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix} \quad A = \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix} \quad b = \begin{pmatrix} c \\ 0 \end{pmatrix}$$

Everything about dynamics and stability is controlled by eigenvalues of A

Ignore constant term B , to study deviation from steady state: $X_{t+1} = AX_t$

Try solution: $X_t = v b^t$ plug in $\Rightarrow v b^{t+1} = A(v b^t) \Rightarrow b v = A v$

Here b is an eigenvalue of A , with eigenvector v .

A nonzero v exists iff $\det(A - bI) = 0$

Characteristic equation: $p(b) = |A - bI| = 0$. For a 2×2 matrix:

$$p(b) = b^2 - Tb + D = 0. \quad \text{where } T = \text{trace}(A), D = \det(A)$$

$$T = b_1 + b_2 \quad D = b_1 \cdot b_2 \Rightarrow p(b) = b^2 - (b_1 + b_2)b + b_1 b_2 = (b - b_1)(b - b_2)$$

eigenvalues are the solution of the characteristic equation: b_1, b_2 .

Back to the example, since $A = \begin{pmatrix} -a_1 & -a_2 \\ 1 & 0 \end{pmatrix} \Rightarrow \begin{cases} T = \text{tr}(A) = -a_1 \\ D = \det(A) = a_2 \end{cases}$

Thus, characteristic equation: $b^2 + a_1 b + a_2 = 0$.

eigenvalues, $\frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2}$ if $a_1^2 > 4a_2$.

Stability triangle on the (T-D plane)

1° $\Delta = T^2 - 4D$

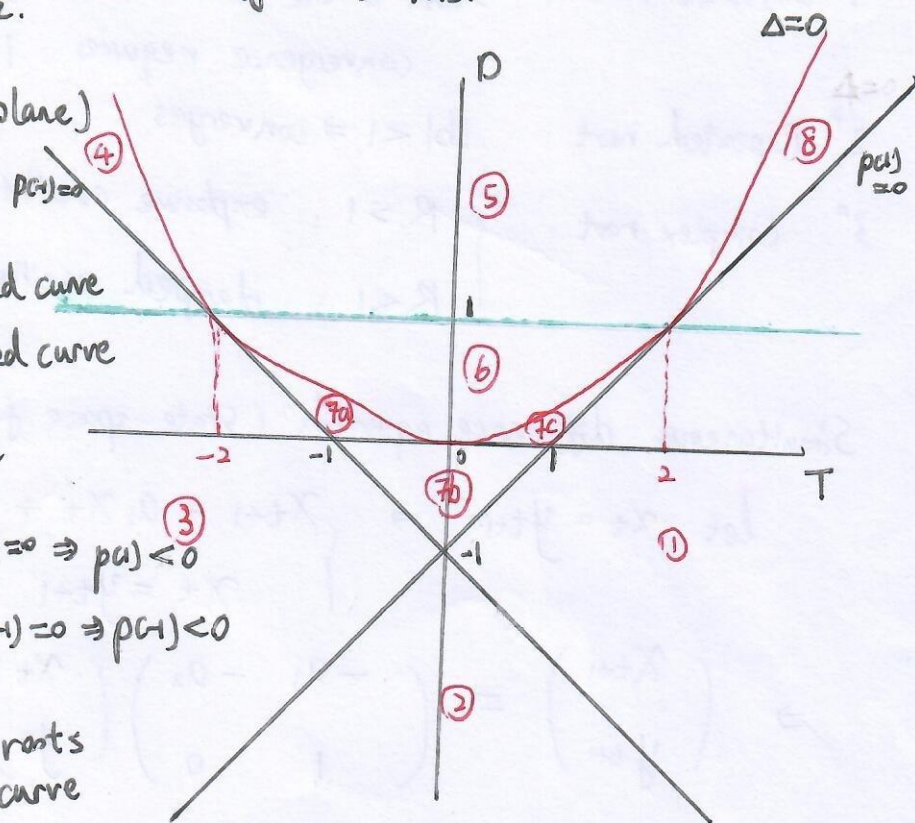
$\Delta < 0$ complex roots $\rightarrow$ above red curve
 $\Delta > 0$ real roots $\rightarrow$ below red curve

2° $p(1) = 0$ $p(-1) = 0$ black curve

$p(1) = 1 - T + D$, right to $p(1) = 0 \Rightarrow p(1) < 0$

$p(-1) = 1 + T + D$, left to $p(-1) = 0 \Rightarrow p(-1) < 0$

3° $D = 1$ and $|b| = 1$ for complex roots
 green curve



Why D controls $|b|$ in complex case

If roots are complex conjugates $b_{1,2} = r e^{\pm i\theta} \Rightarrow D = b_1 b_2 = r^2 \Rightarrow r = \sqrt{D}$

stable ($r < 1$) $\Rightarrow D < 1$ Thus, above green curve $|b| > 1$
 below green curve $|b| < 1$

★ Conditions for two stable roots (both inside unit circle)

$D < 1$ and $p(1) > 0$ and $p(-1) > 0 \Rightarrow D < 1$ $1 - T + D > 0$ $1 + T + D > 0$

$\Rightarrow D < 1$ and $|T| < 1 + D$

complex roots with modulus < 1 , if $\Delta < 0$ ($T^2 < 4D$)

real roots both between -1 and 1 , if $\Delta > 0$ ($T^2 > 4D$)

★ Conditions for one stable root and one unstable root.

$$\begin{aligned} & p(1) < 0 \& p(-1) > 0 \quad \text{or} \quad p(1) > 0 \& p(-1) < 0. \\ & \begin{cases} 1-T+D < 0 \\ 1+T+D > 0 \end{cases} \quad \text{or} \quad \begin{cases} 1-T+D > 0 \\ 1+T+D < 0 \end{cases} \end{aligned} \quad \text{Thus, } |T| > |1+D|$$

$p(1) \& p(-1)$ has opposite signs $\Rightarrow$ There is a root in the interval $(-1, 1)$

$p(1) < 0 \& p(b) \rightarrow +\infty$, as $b \rightarrow +\infty$ / $p(-1) < 0 \& p(b) \rightarrow +\infty$, as $b \rightarrow -\infty$ $\Rightarrow$ There is another root outside the interval $(-1, 1)$

★ Conditions for explosive and unstable roots (both roots outside unit circle or complex with modulus > 1)

Four cases

1° $p(1) < 0$, $p(-1) < 0$, $D < 0$, $|T| > 1$

2° $p(1) > 0$, $p(-1) > 0$, $D > 1$, $T < -2$

3° $\Delta < 0$ and $D > 1$

4° $p(1) > 0$, $p(-1) > 0$, $D > 1$, $T > 2$.

★ Region-by-region summary.

Region	Conditions	Roots
①	$p(1) < 0 \& p(-1) > 0$.	saddle $\begin{cases} 1 \text{ stable root in } (-1, 1) \\ 1 \text{ unstable root in } (1, +\infty) \end{cases}$
②	$p(1) < 0 \& p(-1) < 0$	explosive $\begin{cases} 1 \text{ unstable root in } (1, +\infty) \\ 1 \text{ unstable root in } (-\infty, -1) \end{cases}$
③	$p(1) > 0 \& p(-1) < 0$	saddle $\begin{cases} 1 \text{ stable root in } (-1, 1) \\ 1 \text{ unstable root in } (-\infty, -1) \end{cases}$
④	$p(1) > 0 \& p(-1) > 0$ & $\Delta > 0 \& D > 1$ ($T < -2$)	explosive : 2 real unstable roots in $(-\infty, -1)$
⑤	$p(1) > 0 \& p(-1) > 0$ & $\Delta < 0 \& D > 1$	explosive : complex conjugates with modulus $r = \sqrt{D} > 1$, oscillates & diverge
⑥	$p(1) > 0 \& p(-1) > 0$ & $\Delta < 0 \& D < 1$ ($-2 < T < 2$)	stable : complex conjugates with modulus $r = \sqrt{D} < 1$, oscillates & converge
⑦	$\Delta > 0 \& D < 1 \& p(1) > 0 \& p(-1) > 0$	stable : 2 real roots $ b_1 < 1$ $ b_2 < 1$
⑦a	$D > 0 \& T < 0$ (same sign & $T = b_1 + b_2$ negative)	$-1 < b_2 \leq b_1 < 0$ decay but can flip sign
⑦b	$D < 0$ (opposite sign $D = b_1 b_2$)	$-1 < b_2 < 0 < b_1 < 1$ can flip sign
⑦c	$D > 0 \& T > 0$ (same sign & $T = b_1 + b_2$ positive)	$0 < b_2 \leq b_1 < 1$ monotone converge

(8)	$\Delta > 0, D > 1, T > 2$ $p(1) > 0 \quad p(-1) > 0$	explosive: 2 real roots $b_1 > 1$ $b_2 > 1$	(8)
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6) Blanchard and Kahn (BK) Conditions

Predetermined variable: initial value is given, states like k, h, b capital hour bond

Jump variable: initial value not given, can jump to satisfy equilibrium like c, p .

Stable roots = # predetermined variables $\Rightarrow$ Saddle path (Determinacy)

Stable roots < # predetermined variables $\Rightarrow$ Source (Explosive, no convergent equilibrium)

Stable roots > # predetermined variables $\Rightarrow$ Sink (Indeterminacy)

Why?

Linearize around steady state, we write deviations as a first-order system:

$$X_{t+1} = AX_t$$

If A has eigenpairs (b_i, v_i) the general solution can be written as sum of modes,

$$X_t = \sum_i \alpha_i b_i^t v_i$$

if $|b_i| < 1$, mode dies out $\Rightarrow$ stable

if $|b_i| > 1$, mode blows up $\Rightarrow$ unstable/explosive.

* A convergent equilibrium path must impose: $\alpha_i = 0$ for every $|b_i| > 1$
 which is a set of restrictions, one restriction per unstable root.

Now partition variables into $\begin{cases} \text{predetermined states } s_t \text{ (given } s_0) \text{ with dimension } n_s \\ \text{jump variables } j_t \text{ (free } j_0) \text{ with dimension } n_j \end{cases}$

$X_t = (s_t, j_t)$ initial conditions pin down s_0 , but can choose j_0 .

let $n_u = \#$ unstable roots $n_l = \#$ stable roots

$n_u + n_l = n_s + n_j$ to satisfy n_u restrictions, we have n_j free choices to satisfy them. Thus $n_u = n_j \Leftrightarrow n_l = n_s$.

Case A: $n_l = n_s$ ($n_u = n_j$): jump variables jump to land on saddle path.

Case B: $n_l < n_s$ ($n_u > n_j$): too many unstable modes to eliminate, but too few jump variables to adjust.

Case C: $n_l > n_s$ ($n_u < n_j$): more jump freedom needed to kill unstable modes. (multiple equilibria)